1 Driving factors

This document shows the summary of candidate driving factors of **the number of potential ride-hailing usage opportunities(M)** and **probability of choosing robotaxi for that usage opportunity** derived from the documents of `02_demand_variables` and `03_city_entry_conditions`.

A. Driving factors of M

The papers below mainly explain **observed ride-hailing usage**, so they do not include the 'unrealized demand' included in the definition of M, making direct estimation impossible. However, I suppose the demand patterns provided by these papers lead to constructing 'potential travel opportunities M' through modeling.

Variable	Supporting paper	Scope established in the paper	Measurement data
Population and population density	Elkhouly and Alhadidi, *Ride-sharing Determinants* (paper); Zhang et al., *Examining spatial heterogeneity⋯* (paper)	Relationship of population and local density with conventional ride-hailing usage. Density effects may vary by area	ACS
Employment and economic activity	Elkhouly and Alhadidi (paper)	Uses the economically active population and employment environment to explain usage	LEHD/LODES, ACS
Income	Elkhouly and Alhadidi (paper)	Analyzes the relationship between income and usage in Chicago	ACS B19013
Share of car-free households	Elkhouly and Alhadidi (paper)	Analyzes the relationship between households without cars and usage	ACS B08201
Land-use mix, accessibility, and commercial activity	Zhang et al., *Examining spatial heterogeneity⋯* (paper)	Nonlinear and spatial differences in the urban environment and ridesourcing usage	EPA SLD
Restaurant and leisure-facility density	Loa, Lee, and Circella, *Just another ride-hailing?* (paper)	Relationship with **robotaxi trip counts in SF**. Does not separate the effects of M and p	Census CBP, OpenStreetMap
Public transport and walking environment	Zhang et al. (paper); Cats et al., *Beyond the dichotomy⋯* (paper)	Analyzes the local environment and travel-time competitiveness and substitutability with public transport. Not a uniform increase/decrease effect	EPA SLD, GTFS
Airport-related market size	Zhang et al. (paper)	Analyzes airport travel as a separate spatial category. **Does not estimate the direct effect of airport passenger counts**	FAA passenger data. Used as a proxy for market size
Time, day of week, and season	Elkhouly and Alhadidi (paper)	Analyzes differences in usage over time. Specific variables such as holidays are selected through separate design	Dates and times in trip data
Weather	Additional: Liu, Jiang, and Chen, *Quantifying the impact of weather on ride-hailing ridership: Evidence from Haikou, China* (2021, paper)	Analyzes the relationship between weather and ride-hailing usage. Effects in U.S. cities require revalidation	NOAA CDO

B. Driving factors of p

Variable	Supporting paper	Scope of application	Required measurement data
Relative fare	Yin and Cherchi, *Willingness to pay for automated taxis* (paper)	Choice experiment between conventional and automated taxis. Examines responses to fare differences. Own-service fare depends on operator decisions.	Fares of competing services + choice outcomes
Relative waiting time	Yin and Cherchi (paper)	Examines responses to waiting-time differences. Own-service waiting time depends on fleet size and allocation.	Waiting times of competing services + choice outcomes
In-vehicle travel time	Yin and Cherchi (paper)	Examines trade-offs between fare and travel time. Distinguish time affected by operating policies from underlying traffic conditions.	Travel times of competing services + choice outcomes
Trust, price value, and usage habits	Wei et al., *Using the Extended Unified Theory* (paper); Yin, Cherchi, and Nettle (2023)	Evidence on acceptance intentions and behavior related to price value, habits, and experience. Wei et al. do not include trust. Survey scores are not choice probabilities.	Individual attitude and habit surveys; choice and usage data
Age and robotaxi-related knowledge	Yin and Cherchi (paper); Wei et al. (paper)	Choice and acceptance differences by individual characteristics. Do not impose a uniform sign on age. Distinguish general AV knowledge from awareness of actual service.	Individual age, knowledge, and service awareness linked to choice data
User reviews and in-vehicle support features	Yin and Cherchi (paper)	Examines reviews, destination changes, and operator contact. Support features are potential operator decisions; reviews also reflect past usage.	Reviews and service features + choice outcomes
Experience using robotaxis	Yin, Cherchi, and Nettle, *Living in a City Where Automated Taxis are Operating and Using Them* (paper)	Relates preferences to residence in an operating area and individual usage experience. A uniform positive experience effect is not established.	Individual experience before the decision; service-operation records
Competitiveness of alternatives	Cats et al., *Beyond the dichotomy* (paper)	Evidence from conventional ride-hailing. Substitution and complementarity depend on travel-time competitiveness with alternatives.	Alternative availability, fares, and travel times; GTFS, NTD
Share of car-free households	Elkhouly and Alhadidi, *Ride-sharing Determinants* (paper); Loa, Lee, and Circella, *Just another ride-hailing?* (paper)	Positive associations with conventional ride-hailing usage in Chicago and robotaxi trip origins in SF. Candidate measure of limited access to a private-car alternative.	ACS B08201: car-free households / all households
Income	Elkhouly and Alhadidi; Loa, Lee, and Circella; Yin and Cherchi; Wei et al.	Positive associations in aggregate usage data from Chicago and SF. Individual choice experiments and surveys do not consistently establish the same effect.	ACS B19013; income-group shares if needed
Employment and economic activity	Elkhouly and Alhadidi (paper); Loa, Lee, and Circella (paper)	Economic activity share is positively associated with conventional ride-hailing usage in Chicago; employment density with robotaxi trip destinations in SF. Distinguish the two measures.	LEHD/LODES; ACS B23025
Population density	Elkhouly and Alhadidi; Zhang et al., *Examining spatial heterogeneity* (paper); Loa, Lee, and Circella	Spatially varying relationships for conventional ride-hailing. Negative association with robotaxi usage in SF. Effects at the city level require separate validation.	ACS population + area within the same boundary

Land-use mix, accessibility, and commercial activity	Zhang et al. (paper); Loa, Lee, and Circella (paper)	Nonlinear and spatially heterogeneous relationships for conventional ride-hailing. Positive association with land-use mix in SF. Select a representative measure and aggregation method.	EPA Smart Location Database
Public transport and walking environment	Zhang et al.; Cats et al. (paper); Loa, Lee, and Circella	Substitution and complementarity coexist for conventional ride-hailing. Positive associations of transit frequency and walkability with robotaxi usage in SF.	EPA SLD; GTFS; NTD
Restaurant and leisure-facility density	Loa, Lee, and Circella, *Just another ride-hailing?* (paper)	Positive association with robotaxi trip counts in SF. The evidence does not separate the effects of M and p.	Census CBP; OpenStreetMap
Age composition	Loa, Lee, and Circella; Etminani-Ghasrodashti et al. (2022) (paper)	Aggregate and pilot evidence concerning younger users. Individual age effects do not directly establish effects of an area's age composition.	ACS B01001; initially consider the share aged 18–34
Student share	Etminani-Ghasrodashti et al. (2022) (paper)	Actual usage evidence from the Arlington campus pilot. Account for free student rides and the campus setting.	ACS enrollment data; local student population; fare-support conditions
Education and technology affinity	Krueger, Rashidi, and Rose (2016); Haboucha et al. (2017); Gkartzonikas and Gkritza (2019)	Additional candidates considered in SAV choice experiments and reviews. Education and technology affinity are distinct; specific findings require confirmation.	ACS education data; technology-affinity surveys
Existing ride-hailing habits	Wei et al. (paper); conventional ride-hailing studies	Examines relationships of habits and previous usage with adoption. Past trip counts are a proxy for habits, not the same measure as a habit scale.	Common habit surveys; conventional ride-hailing usage before the decision
Safety perceptions and exposure	SAV choice experiments and reviews; bibliographic details and findings of Khan et al. (2022) require confirmation	Distinguish perceived safety from actual crash or crime exposure. Post-use satisfaction should not directly substitute for safety perceptions before the decision.	Common safety surveys; crash, crime, and exposure data
Regulatory permission	Permits and operating requirements from the relevant authorities; NCSL legislative information	An operating-feasibility condition. Treat areas without the required permission separately in entry feasibility.	Valid permits and operating requirements by area
Climate and ODD	Weather records and operators' Operational Design Domain information	Climate is an exogenous condition. ODD and service availability depend on technical and operating conditions; distinguish these from preference effects.	NOAA; operator ODD; operating-restriction records
Time since service launch	Yin, Cherchi, and Nettle (2023) (paper); service-launch and operation records	Candidate proxy for accumulated awareness and experience. Launch timing depends on operator decisions; the effect of elapsed time requires separate validation.	Company announcements; permit and launch dates; annual operation records

Measurement principle. Robotaxi trips and M must use the same geographic boundary, year, and trip unit. If the numerator contains completed trips, p measures a realized usage share affected by supply conditions.